arXiv:2203.01556v2 [cs.HC] 15 Jun 2022

DareFightingICE Competition: A Fighting Game Sound Design and AI Competition

Ibrahim Khan, Thai Van Nguyen, Xincheng Dai
Graduate School of Information Science and Engineering
Ritsumeikan University
Kusatsu, Shiga, Japan
{gr0556vx, gr0557fv, gr0502pv}@ed.ritsumei.ac.jp

Ruck Thawonmas
College of Information Science and Engineering
Ritsumeikan University
Kusatsu, Shiga, Japan
ruck@is.ritsumei.ac.jp

***Abstract*—This paper presents a new competition – at the 2022 IEEE Conference on Games (CoG) – called DareFightingICE Competition. The competition has two tracks: a sound design track and an AI track. The game platform for this competition is also called DareFightingICE, a fighting game platform. DareFightingICE is a sound-design-enhanced version of FightingICE, used earlier in a competition at CoG until 2021 to promote artificial intelligence (AI) research in fighting games. In the sound design track, participants compete for the best sound design, given the default sound design of DareFightingICE as a sample, where we define a sound design as a set of sound effects combined with the source code that implements their timing-control algorithm. Participants of the AI track are asked to develop their AI algorithm that controls a character given only sound as the input (blind AI) to fight against their opponent; a sample deep-learning blind AI will be provided by us. Our means to maximize the synergy between the two tracks are also described. This competition serves to come up with effective sound designs for visually impaired players, a group in the gaming community which has been mostly ignored. To the best of our knowledge, DareFightingICE Competition is the first of its kind within and outside of CoG.**

***Index Terms*—Sound Design Competition, AI Competition, Visually Impaired Players, DareFightingICE, FightingICE, Fighting Game**

I. Introduction

Video games are on the rise, with the global market revenue of video games being increased from 178.37 million to 197.11 million (U.S dollar) from 2021 to 2022 [1]. Their popularity is increasing at a fast and steady pace. Since the start, video games have been a testing ground for Artificial Intelligence (AI) [2]. There have been many breakthroughs in AI stemming from the research on video games [3]. The rise in the popularity of games is the reason that game developers are looking to expand their audience. One of which is visually impaired players (VI Players), which have been mostly ignored in the past [4]. Game developers or researchers are adding new features such as specific audio cues so that VI Players can also experience and enjoy the games [5].

Our focus is on fighting games. In these fighting games, players go against another player or a computer player in a one-versus-one fight, using different attacks and abilities to overcome the opponent. These fighting games are mostly two-dimensional, which means that the players can only move in two dimensions. In addition, because fighting games are fairly simple compared to other genres of video games when they are given the right sound designs, VI Players will have a much easier time playing than other genres; in this work, we define a sound design as a set of sound effects combined with the source code that implements their timing-control algorithm.

Our goal is to use a fighting game platform called "DareFightingICE", which is "FightingICE" [6] with an enhanced sound design, to help VI Players play the game and to set the DareFightingICE platform as a testing ground for AI algorithms that only use sound as the input. We plan to achieve this goal through running a competition in the upcoming IEEE Conference on Games (CoG). This competition has two tracks: one is a sound design competition, and the other is an AI competition that only uses sound as the input.

We anticipate that this research will give voice to one of the most ignored members of the gaming community – VI Players – and that we hope the competition will give hints to the production of sound designs that can be used in video games to make them compatible with VI Players. Along with the sound design, this competition is going to open doors to a new wave of blind AI algorithms, i.e., those that only use sound as the input. Our contributions are as follows: first, we provide a platform for developers and researchers to create sound designs considering VI Players for fighting games; second, we are opening the door for research in AI algorithms that only use sound as the input; and third, the two tracks – Sound Design and AI – in this competition can be expected to have synergy between respective research areas.

II. Related Work

Since our research is a combination of three different topics (sound designs in video games, games for VI Players, and competitions), this section is divided into three parts.

A. Sound/ Sound Designs in Video Games

There has been plenty of research on the importance of sound in video games. It has been established that music played in video games can affect a person's daily life, personality, and such [7]. These effects can be formed because of a person's attachment to the music used or played in a game. Sound in video games in terms of voiceover (audio dialogue played by the player or the non-playable characters)

arXiv:2203.01556v2 [cs.HC] 15 Jun 2022

DareFightingICE竞赛：一项格斗游戏声音设计与人工智能竞赛

Ibrahim Khan、Thai Van Nguyen、
Xincheng Dai立命馆大学信息科学与工程研究生院日本滋贺草津{gr0556vx,gr0557fv, gr0502pv}
@ed.ritsumei.ac.jp

Ruck Thawonmas立命馆大学信息科学与工程学院日本滋贺草津
ruck@is.ritsumei.ac.jp

摘要——本文介绍了在2022年IEEE游戏会议（CoG）上举办的一项新竞赛——DareFightingICE竞赛。该竞赛设有两个赛道：声音设计赛道和人工智能赛道。本竞赛采用的游戏平台同样名为DareFightingICE，是一个格斗游戏平台。DareFightingICE是FightingICE的声音设计增强版；截至2021年，FightingICE曾用于CoG的一项竞赛，以推动格斗游戏中的人工智能（AI）研究。在声音设计赛道中，参赛者以DareFightingICE的默认声音设计为示例，角逐最佳声音设计。我们将声音设计定义为一组音效及实现其时序控制算法的源代码。人工智能赛道的参赛者需要开发仅以声音为输入、控制角色与对手交战的人工智能算法（盲人 AI）；我们将提供一个基于深度学习的盲打人工智能示例。本文还介绍了我们为最大限度发挥两个赛道之间的协同作用而采取的方法。本竞赛旨在为视障玩家设计有效的声音方案，这一群体在游戏社区中一直鲜受关注。据我们所知，无论在CoG内部还是外部，DareFightingICE竞赛都是首个此类竞赛。

索引词——**声音设计竞赛、人工智能竞赛、视障玩家、DareFightingICE、FightingICE、格斗游戏**

I. 引言

电子游戏正蓬勃发展，其全球市场收入从2021年的1.7837亿美元增长至 2022 [1]的1.9711亿美元。电子游戏的受欢迎程度正以快速而稳定的步伐提升。自诞生以来，电子游戏一直是人工智能（AI）的试验场 [2]。人工智能领域的许多突破都源于对电子游戏的研究 [3]。游戏日益普及，促使游戏开发者寻求扩大受众群体。其中一个群体便是视障玩家，他们在过去大多被忽视 [4]。游戏开发者或研究人员正在添加特定音频提示等新功能，让视障玩家也能体验并享受游戏 [5]。

我们关注的是格斗游戏。在这类格斗游戏中，玩家与另一名玩家或电脑玩家进行一对一对战，使用不同的攻击招式和能力击败对手。这些格斗游戏大多是二维的，这意味着玩家只能在两个维度上移动。此外，由于格斗游戏相当简单，相较于其他类型的电子游戏，只要配备恰当的声音设计，视障玩家玩起来就会比玩其他类型的游戏轻松得多；在本研究中，我们将声音设计定义为一组音效及实现其时序控制算法的源代码。

我们的目标是使用名为"Dare- FightingICE"的格斗游戏平台，帮助视障玩家游玩，并将DareFightingICE平台打造为仅以声音作为输入的人工智能算法的试验场。该平台是在"FightingICE" [6] 的基础上增强声音设计而成的。我们计划通过在即将召开的IEEE游戏大会（CoG）上举办竞赛来实现这一目标。本次竞赛设有两个赛道：一个是声音设计竞赛，另一个是仅以声音作为输入的人工智能竞赛。

我们期望这项研究能让游戏社区中最受忽视的群体之一——视障玩家——发出自己的声音，也希望本次竞赛能为制作适用于电子游戏的声音设计提供启示，使电子游戏能够满足视障玩家的需求。除声音设计外，本次竞赛还将为新一代盲人工智能算法，即仅以声音作为输入的算法，开辟道路。我们的贡献如下：首先，我们为开发者和研究人员提供了一个平台，使其能够面向视障玩家为格斗游戏创建声音设计；其次，我们为仅以声音作为输入的人工智能算法研究开辟了道路；第三，本次竞赛的声音设计和人工智能两个赛道有望促进各自研究领域之间的协同。

II. 相关工作

由于我们的研究结合了三个不同的主题（电子游戏中的声音设计、面向视障玩家的游戏以及竞赛），本节分为三个部分。

A. 电子游戏中的声音／声音设计

关于声音在电子游戏中的重要性，已有大量研究。研究已证实，电子游戏中播放的音乐会影响一个人的日常生活、个性等方面 [7]。这些影响可能源于人们对游戏中使用或播放的音乐所产生的情感依恋。电子游戏中的配音（由玩家或非玩家角色说出的语音对话）

has been known to be more engaging for the players [8]. These voiceovers also help in remembering the information provided to the players in the game. The background music being played in the game also has an important role to play. It was observed that players performed better with background music than those who played the game without background music [9].

Similarly, it has been observed that the music played in video games must meet the mood or tone of the game to be engaging for the players [10]. A bad example of this would be to play calm music when the game is at a climax. Moving from general music to specific sound effects and audio cues, it has been seen that directional or 3D sound effects help the players know more about their environment, such as knowing the position of other players [11]. These 3D sound effects include sound like footsteps and other movement sound.

So far, this sub-section has been about sound in video games. Now shifting from sound in video games to sound designs, there has been only a little research when it comes to sound designs in video games. And even fewer have had positive results that can be useful in the creation of a sound design. Scarce as the research may be, sound designs have been proposed for game genres such as First Person Shooter and Real-Time Strategy [12]. These two sound designs have specific sound effects for respective game genres.

B. Games for VI Players and Fighting Games

This sub-section includes research related to VI Players and fighting games that have some sort of support for such players. The reason behind the inclusion of fighting games here is because the main platform being used for our research is a fighting game platform and thereby our focus is on the fighting game genre of video games. Before making games for VI Players, it is important to know what these players want in a game and what they look forward to. Keeping this in mind it was observed that fighting games are the second most played games among VI Players [13]. There has been some research on serious games (a game with a purpose like an educational game) which involve VI Players. It was observed that audio cues in a serious game were able to convey spatial information about a room to such players [14]. These audio cues were played as VI Players explored a room in the game.

Moving from research on games for VI Players to fighting games with VI Players compatibility i.e., fighting games that enable the VI player to play with their sound effects. In this work, we focus on three fighting games with such compatibility, to the best of our knowledge. The first, but the least compatible among the three, is Takken 7, the latest game in the game series called Takken. This game has specific sound for different characters making it easy to recognize and remember. The next is Mortal Kombat 11, the latest game in the fighting game series Mortal Kombat. This game is even more helpful in terms of VI Players compatibility than Takken 7 because not only does it have specific sound for different characters, but it also gives different audio cues for different characters on the character selection screen. Lastly, Killer Instinct is a fighting game that has stereo sound and different sound effects, such as a sound effect played when the player reaches the end of the stage, and a sound effect indicating the increase in energy, to help VI Players play the game. Different concepts of audio cues in this game are adopted in DareFightingICE.

All the aforementioned commercial games can be played by VI Players, but there is room for improvement. The Tekken and Mortal Kombat series do not have 3D sound, and Killer Instinct's 3D sound can get VI Players overwhelmed by the different sound effects of special moves a character can perform, which are loud. In addition, these games are somewhat not user-friendly for VI Players, as they were not specifically designed for VI Players or at least their publishers did not claim so. Our new fighting game platform is created to solve all these problems.

C. Competitions and Sound Design Competitions

This sub-section is about different competitions regarding video games at CoG as well as sound design competitions. CoG has many competitions. Those with recent papers at CoG include Hanabi Competition [15] which is based on the game Hanabi, a cooperative card game; Blood Bowl Competition [16] which is a board game competition for AI; VGC AI Competition [17] which is an AI competition based on Pokémon; and Carle's Game AI competition [18] which is a challenge in open-ended machine exploration and creativity.

Now let us shift from competitions in CoG to sound design competitions outside of CoG. This is because so far there has been no sound design competition in CoG. There have been a few sound design competitions conducted in the past. Game Music and Sound Design Conference (GMSDC'18) is a sound design competition for which the participants are allowed to submit a sound design for all genres of games [19]. The Berlin International Sound Design Competition (BISDC'18) is a sound design competition for game developers and filmmakers, allowing sound designs for all genres of games and films [20]. Film Music Contest (FMC)'s Music for Video Game (MVG'21) is a sound design competition for all types of games, just like GMSDC'18 [21]. These three sound design competitions are conducted outside of any academic conference related to games.

Table I summarizes the contributions of nine competitions in CoG 21 and the aforementioned three sound design competitions. To the best of our knowledge, those nine were the competitions at CoG 21 that had participants, while the other three were the only prominent sound design competitions in recent years. The definition of each contribution in the table is as follows:

Normal AI: Targeting AI algorithms that can access game state data in all kinds of forms.

Sound Design: Targeting general sound designs for any field or game, with no requirements or restrictions on what should be included.

Blind AI: Targeting AI algorithms that only use sound as the input.

已被证实能让玩家更加投入 [8]。这些配音还有助于玩家记住游戏中提供的信息。游戏中播放的背景音乐也发挥着重要作用。研究发现，与在没有背景音乐的情况下玩游戏的玩家相比，有背景音乐时玩家的表现更好 [9]。

同样，有研究发现，电子游戏中播放的音乐必须契合游戏的氛围或基调，才能吸引玩家 [10]。一个反面例子是在游戏达到高潮时播放舒缓的音乐。从一般的音乐转向具体的音效和音频提示，研究发现，定向或三维音效有助于玩家更好地了解周围环境，例如获知其他玩家的位置 [11]。这些三维音效包括脚步声和其他移动时发出的声音。

到目前为止，本小节讨论的都是电子游戏中的声音。现在将讨论重点从电子游戏中的声音转向声音设计，目前关于电子游戏声音设计的研究还很少。其中，取得可用于声音设计创作的积极成果的研究更是少之又少。尽管研究不多，但已有研究针对第一人称射击游戏和即时战略游戏等游戏类型提出了声音设计方案 [12]。这两种声音设计针对各自的游戏类型采用了特定的音效。

B. 面向视障玩家的游戏与格斗游戏

本小节介绍与视障玩家相关的研究，以及为此类玩家提供某种支持的格斗游戏。之所以在此纳入格斗游戏，是因为本研究使用的主要平台是格斗游戏平台，因此我们关注的是电子游戏中的格斗游戏类型。在为视障玩家开发游戏之前，了解这些玩家希望游戏具备哪些内容，以及他们对游戏有何期待，十分重要。基于这一考虑，我们发现，格斗游戏在视障玩家最常玩的游戏类型中排名第二 [13]。已有一些关于严肃游戏（具有特定目的的游戏，如教育游戏）的研究涉及视障玩家。研究发现，严肃游戏中的音频提示能够向此类玩家传达房间的空间信息 [14]。当视障玩家在游戏中探索房间时，系统会播放这些音频提示。

接下来，我们从面向视障玩家的游戏研究转向具有视障玩家兼容性的格斗游戏，即能够让视障玩家借助音效游玩的格斗游戏。在本研究中，我们重点关注据我们所知具有这种兼容性的三款格斗游戏。第一款是铁拳系列的最新作铁拳7，但它在三者中的兼容性最低。这款游戏为不同角色配备了专属声音，便于识别和记忆。第二款是真人快打格斗游戏系列的最新作真人快打11。这款游戏在视障玩家兼容性方面比铁拳7更出色，因为它不仅为不同角色配备了专属声音，还在角色选择界面为不同角色提供了不同的音频提示。最后，杀手本能是一款格斗游戏，具备立体声和多种音效，例如玩家到达场地边缘时播放的音效，以及提示能量增加的音效，以帮助视障玩家游玩。DareFightingICE借鉴了这款游戏中多种音频提示的设计理念。

视障玩家可以游玩上述所有商业游戏，但这些游戏仍有改进空间。《铁拳》和《真人快打》系列没有三维音效，而《杀手本能》的三维音效可能会让视障玩家难以应付，因为角色施展各种特殊招式时会产生不同且响亮的音效。此外，这些游戏对视障玩家不够友好，因为它们并非专为视障玩家设计，或者至少其发行商并未作出这样的声明。我们全新的格斗游戏平台旨在解决所有这些问题。

C. 竞赛与声音设计竞赛

本小节介绍CoG中与电子游戏相关的各类竞赛，以及声音设计竞赛。CoG设有众多竞赛。近期在CoG上发表过相关论文的竞赛包括：花火竞赛 [15] ，该竞赛基于合作卡牌游戏花火；血碗竞赛 [16]，这是一项面向人工智能的桌面游戏竞赛；VGC人工智能竞赛 [17] ，这是一项基于宝可梦的人工智能竞赛；以及Carle游戏人工智能竞赛 [18] ，这是一项涉及开放式机器探索与创造力的挑战。

现在，我们将讨论重点从CoG中的竞赛转向CoG之外的声音设计竞赛。这是因为迄今为止，CoG尚未举办过声音设计竞赛。过去曾举办过一些声音设计竞赛。游戏音乐与声音设计大会（GMSDC’18）是一项声音设计竞赛，参赛者可以提交适用于所有游戏类型的声音设计 [19]。柏林国际声音设计竞赛（BISDC’18）是一项面向游戏开发者和电影制作人的声音设计竞赛，接受适用于所有游戏和电影类型的声音设计[20]。电影音乐大赛（FMC）的电子游戏音乐（MVG’21）是一项面向所有游戏类型的声音设计竞赛，与GMSDC’18 [21]类似。这三项声音设计竞赛均不依托任何游戏相关学术会议举办。

表 I 总结了 CoG 21 的九项竞赛以及上述三项声音设计竞赛的贡献。据我们所知，这九项竞赛是 CoG 21 中有参赛者参与的竞赛，而另外三项则是近年来仅有的知名声音设计竞赛。表中各项贡献的定义如下：

常规人工智能： 面向能够访问各种形式的游戏状态数据的人工智能算法。**声音设计：** 面向任何领域或游戏的通用声音设计，对应包含的内容不作任何要求或限制。**盲人 AI：** 面向仅使用声音作为输入的人工智能算法。

TABLE I
COMPARISON OF EXISTING COMPETITIONS AND OUR COMPETITION OVER FOUR DIFFERENT CONTRIBUTION POINTS. (A) NORMAL AI CONTRIBUTION (B) SOUND DESIGN CONTRIBUTION (C) BLIND AI CONTRIBUTION (D) SOUND DESIGN FOR VI PLAYERS

Competition Names	Contributions			
	A	*B*	*C*	*D*
Bot Bowl III	√	×	×	×
ColorShapeLinks	√	×	×	×
Dota 2 5v5 AI Competition	√	×	×	×
Fighting Game AI Competition	√	×	×	×
GVGAI single-player learning competition	√	×	×	×
Ludii AI Competition	√	×	×	×
StarCraft AI Competition	√	×	×	×
Strategy Card Game AI Competition	√	×	×	×
AI Snakes Game	√	×	×	×
GMSDC'18	×	√	×	×
BISDC'18	×	√	×	×
MVG'21	×	√	×	×
Our Competition	×	×	√	√

√ : Yes , × : No

Sound Design for VI Players: Targeting sound designs for VI Players.

This table should manifest well the position of our competition among the rest. Note that the nine CoG 21 competitions did not provide audio data to their entry AIs.

III. DAREFIGHTINGICE

Compared to its predecessor FightingICE, DareFightingICE has an enhanced sound design, which is made by keeping VI Players in mind. It uses 3D sound effects. Along with the enhanced sound design, DareFightingICE also has a new interface, enabling AIs (henceforth AI in a countable form refers to an AI algorithm) to receive sound data from the game.

The FightingICE platform supports Java programming and is also wrapped for Python programming by Py4J to develop AIs. This platform allows users to create AIs for non-player characters. The platform is also useful for relevant research communities to conduct research as shown in the following paragraph.

In one paper, their authors presented a fighting-game gameplay generation using Monte-Carlo tree search with the evaluation function formed based on highlight cues [22]. In another paper, Chen et al. [23] proposed a method, using FightingICE, to learn utility functions from data collected from human players, to recreate a target behavior. Another paper [24] proposed an algorithm that combines Rolling Horizon Evolution Algorithm with an opponent model using FightingICE; their AI won the 2020 Fighting Game AI Competition. These three are just examples of recent studies utilizing the FightingICE platform.

As with the FightingICE platform, we hope that the DareFightingICE platform is going to open exciting new doors for developers and researchers. With the addition of blind AIs and sound designs for VI Players, DareFightingICE is the first of its kind, as a research platform, for fighting games. We hope to see DareFightingICE being used in research by relevant research and development communities in the near future.

A. Sample Sound Design

Our sample sound design is created and used as the default sound design of DareFightingICE to provide the participants of this competition with a sample of how a sound design can be created. In its current form, there is room for improvement, which is intentional, and it is our hope to see improved and stimulating new sound designs. The sample sound design is created with the java library "lwjgl", which uses openAL.

There is a total of 51 sound effects in the sample sound design including the background music. Some sound effects are the same for similar moves or actions. Since we use openAL, the sound effects are mono, not stereo. This is because to emit 3D sound with openAL the sound effects must be in mono. Of the 51 sound effects, there are three special sound effects (SSEs): Heartbeat, Energy-Increase, and Border-Alert. These SSEs are to help VI Players play the game. Energy-Increase and Border-Alert SSEs are inspired by Killer Instinct, and Heartbeat is from previous research on sound design [12]; their descriptions are as follows: Heartbeat: This sound effect is played when the player's health is below 50. For player 1 the sound effect is played on the left speaker and for player 2 on the right. Energy Increase: This sound effect is played when the player's energy goes +50 from the previous value. For player 1 the sound effect is played on the left speaker and for player 2 on the right. Border Alert: This sound is played when a player reaches the end of the stage on either side. The sound is played on the left side when a player reaches the end of the stage on the left side and the same applies for the right side. Please also use the description environment.

There are three important elements in our design: Listener, Audio-Sources, and Audio-Buffers. Listener, provided by lwjgl, listens to the audio generated by the game. It is responsible for providing the output of audio to the game players. The position of Listener in DareFightingICE is in the middle of the game screen since the camera does not move from its original position. The velocity of Listener is defined by its speed and direction, which is used to create a Doppler effect. The orientation of Listener is represented by two vectors: "at", and "up". The former represents the forward direction, and the latter represents the upward direction.

Audio-Sources are in-game objects or in-game events that produce sound, and in our sample sound design, the position of these Audio-Sources from Listener determines from which direction the players will hear the sound effect. These Audio-Sources also have velocity and orientation. A Doppler effect is created with the help of the velocity of an Audio-Source of interest and the velocity of Listener. Audio-Buffers are all the sound effects that our sample sound design has. Figure 1 shows the provided function to play a sound effect. In this figure, *soundRenderers* contains two renderers (one for rendering sound for the speakers and the other one for AIs), *bufferId* represents the index of an Audio-Buffer, the parameters *x* and

表 I 现有竞赛与本竞赛在四个不同贡献方面的比较。（A）常规人工智能贡献（B）声音设计贡献（C）盲人 AI 贡献（D）面向视障玩家的声音设计

竞赛名称	贡献			
	A	*B*	*C*	*D*
Bot Bowl III	√	×	×	×
ColorShapeLinks	√	×	×	×
Dota 2 5v5 人工智能竞赛	√	×	×	×
格斗游戏人工智能竞赛	√	×	×	×
GVGAI 单人学习竞赛	√	×	×	×
Ludii人工智能竞赛	√	×	×	×
星际争霸人工智能竞赛	√	×	×	×
策略卡牌游戏人工智能竞赛	√	×	×	×
人工智能贪吃蛇游戏	√	×	×	×
GMSDC' 18	×	√	×	×
BISDC' 18	×	√	×	×
MVG' 21	×	√	×	×
我们的竞赛	×	×	√	√

√：是，×：否

面向视障玩家的声音设计：以面向视障玩家的声音设计为目标。

该表应能清楚地展示本竞赛在其他竞赛中的定位。请注意，CoG 21 的九项竞赛均未向参赛人工智能提供音频数据。

III. DAREFIGHTINGICE

与其前身FightingICE相比，DareFightingICE的声音设计得到了增强，并充分考虑了视障玩家的需求。它采用了三维音效。除了增强声音设计外，DareFightingICE还提供了一个新接口，使人工智能（下文中以可数形式使用的“人工智能”指人工智能算法）能够接收游戏中的声音数据。

FightingICE平台支持Java编程，并通过Py4J进行封装，以支持使用Python编程开发人工智能。该平台允许用户为非玩家角色创建人工智能。此外，该平台也有助于相关研究群体开展研究，具体如下段所述。

在一篇论文中，作者提出了一种使用蒙特卡洛树搜索生成格斗游戏玩法的方法，其评估函数基于精彩片段线索构建 [22]。在另一篇论文中，Chen等人 [23] 提出了一种利用FightingICE的方法，从人类玩家收集的数据中学习效用函数，以复现目标行为。还有一篇论文 [24] 提出了一种利用FightingICE将滚动时域进化算法与对手模型相结合的算法；其人工智能赢得了2020年格斗游戏人工智能竞赛。这三项研究只是近期利用FightingICE平台开展研究的几个例子。

与FightingICE平台一样，我们希望Dare- FightingICE平台能为开发者和研究人员开辟令人振奋的新天地。通过引入盲人工智能和面向视障玩家的声音设计，DareFightingICE成为首个此类格斗游戏研究平台。我们希望

在不久的将来，看到相关研发社区将DareFightingICE用于研究。

A. 示例声音设计

我们创建了示例声音设计，并将其用作DareFightingICE 的默认声音设计，为本次竞赛的参赛者提供声音设计的创建示例。该设计目前仍有改进空间，这是有意为之的，我们希望看到经过改进且富有启发性的新声音设计。该示例声音设计使用 Java库“lwjgl”创建，该库使用OpenAL。

示例声音设计共包含 51 种音效，其中包括背景音乐。某些相似的招式或动作使用相同的音效。由于我们使用OpenAL，因此这些音效为单声道，而非立体声。这是因为使用 OpenAL 产生三维音效时，音效必须为单声道。在这 51 种音效中，有三种特殊音效（SSE）：心跳、能量增加和边界警报。这些特殊音效旨在帮助视障玩家进行游戏。能量增加和边界警报特殊音效的灵感来自《杀手本能》，而心跳则源自先前关于声音设计的研究 [12]；具体说明如下：心跳：当玩家的生命值低于 50 时，播放此音效。玩家 1 的音效通过左扬声器播放，玩家 2 的音效通过右扬声器播放。能量增加：当玩家的能量相较于先前数值发生 +50 时，播放此音效。玩家 1 的音效通过左扬声器播放，玩家 2 的音效通过右扬声器播放。边界警报：当玩家到达场地任一侧的边界时，播放此音效。当玩家到达场地左侧边界时，音效从左侧播放，右侧同理。另请使用描述环境。

我们的设计包含三个重要元素：监听器、音源和音频缓冲区。监听器由 lwjgl 提供，用于监听游戏生成的音频，并负责向游戏玩家输出音频。在 DareFightingICE 中，由于摄像机始终保持在初始位置，监听器位于游戏画面的中央。监听器的速度由其速率和方向定义，用于产生多普勒效应。监听器的朝向由两个向量表示：“at”和“up”。前者表示前向方向，后者表示向上方向。

音源是产生声音的游戏内对象或游戏内事件。在我们的示例声音设计中，这些音源相对于监听器的位置决定了玩家将从哪个方向听到音效。这些音源还具有速度和朝向。利用目标音频源的速度和监听器的速度，可以产生多普勒效应。音频缓冲区包含我们的示例声音设计中的所有音效。图1展示了所提供的用于播放音效的函数。在该图中，*soundRenderers* 包含两个渲染器（一个用于为扬声器渲染声音，另一个用于为人工智能渲染声音），*bufferId*表示音频缓冲区的索引，参数*x* 和

```
/**
* Plays an Audio-Source by giving it an
    Audio-Buffer to play and the position where
    to play.
*/
public void play2(AudioSource source,
    AudioBuffer buffer, int x, int y, boolean
    loop){
   for (int i = 0; i < soundRenderers.size();
       i++){
      int sourceId = source.getSourceIds()[i];
      int bufferId = buffer.getBuffers()[i];
      soundRenderers.get(i).play(sourceId,
          bufferId, x, y, loop);
   }
}
```

Fig. 1. Function to play a sound effect

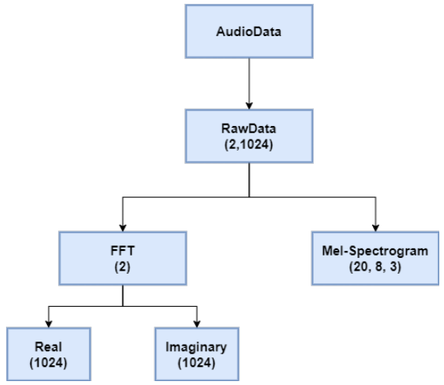


Fig. 2. Sound data structure.

y store the horizontal location and the vertical locations of where to play the sound effect on the stage (map), and *loop* confirms whether to play the sound effect in a loop or not.

B. AI Interface and Blind AIs

Below are details about our AI interface that provides sound data to blind AIs and is used in the competition.

There have been a number of prior studies relating to game playing AIs with sound. Gaina and Stephenson [5] expanded the General Video Game AI framework to support sound and trained an AI that played the game from sound only. Hegde et al. [25] extended the VizDoom framework to provide the in-game sound to AIs and trained them in several scenarios with increasing difficulty to test the perception of sound. Results from these studies show promise in learning to play games from sound. In our project, sound in DareFightingICE will be provided to entry blind AIs so they can be trained to play the game from sound only.

To facilitate AIs with access to sound data, the original interface in FightingICE [6] is extended with a new method *getAudioData*() that provides sound data at each frame of the game. At each frame, sound data are sampled with a length of 16.67 ms; note that DareFightingICE has an FPS value of 60, as in FightingICE. In DareFightingICE, stereo sound is provided in 3D format, which helps effectively represent the game's environment. Therefore, in order to allow AIs' perception of sound that might help them localize in-game objects and in-game events, a two-channel format of sound is provided.

As shown in Fig. 2, not only are raw data provided at each frame, we also provide two transformations of data (Fig. 3): Fast Fourier Transform (FFT) and Mel-Spectrogram (Mel-Spec).

- Raw Data: The raw audio data is a vector $\mathbf{s} \in \mathbf{R^n}, \mathbf{s_i} \in [-\mathbf{1}, \mathbf{1}]$ where n is the number of normalized audio samples. A higher sampling frequency provides more details of the observations. In our work, we perform sampling at a rate of 48000Hz, which allows for high-quality sound and fast sound rendering. In addition, sound data are sampled with a length of 16.67 ms so that our raw audio data consist of 800 samples for each of the two channels: left and right. Note that in our implementation, we perform padding so that raw data of each channel are represented by a 1024-length array for computational efficiency in the following data transformations.
- FFT: We perform the first transformation from raw data to frequency domain using FFT. The output of FFT data consists of two parts: real part and imaginary part. Each part is a 1024-length array.
- Mel-Spec: This type of transformation has gained success in speech processing [26]. Therefore we transform raw audio samples into frequency domain spectrogram with short-term Fourier transform (STFT). STFT is a sequence of Fourier transform of a windowed signal, where the window is moved with a given hop. We then apply Mel scale to the frequencies. We use all the parameters from [26], with a hop size of 10 ms and 80 mel-frequency components, except for the window size. Because the time length of a frame is shorter than 25 ms, we choose the window size of 1024. The output of Mel-Spec is a (80, 3) shaped vector, where 80 is the number of Mel bins and 3 is the number of Mel spectra.

Figure 2 shows the structure of sound data and dimension of each data type provided to the AIs. The number "2" in each dimension represents two channels. We process both left and right audio and store the transformation data into two arrays, each of which stores the output of a single transformation. Figure 3 shows an example of audio wave and transformation of an action using our parameters.

AIs are written in Java or Python and must implement the AIInterface interface shown in Figure 4. The new method $getAudioData()$ provides sound data at each frame of the game. The AI is provided with an $AudioData$ object as the argument to this method. $AudioData$ provides access to information about the aforementioned sound data (Raw Data, FFT, and Mel-Spec).

```
/**
* 通过为音频源提供一个
    要播放的音频缓冲区以及播放
    位置来播放。
*/
public void play2(AudioSource source,
    AudioBuffer buffer,  int x, int y, boolean
    循环){
   for (int i = 0; i < soundRenderers.size();
       i++){
      int sourceId = source.getSourceIds()[i];
      int bufferId = buffer.getBuffers()[i];
      soundRenderers.get(i).play(sourceId,
          bufferId,  x, y,  loop);
   }
}
```

图 1. 播放音效的函数

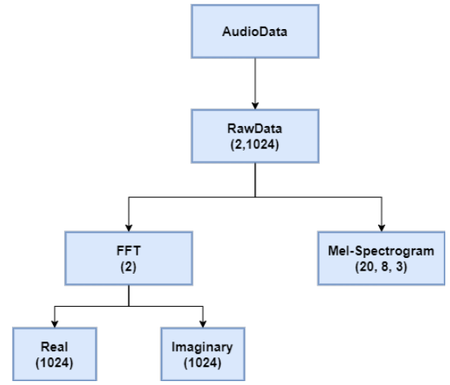


图2. 声音数据结构。

y 存储音效在场景（地图）中播放位置的水平位置和垂直位置，而循环用于确定是否循环播放该音效。

B. 人工智能接口与盲人工智能

以下详细介绍我们在竞赛中使用的人工智能接口，该接口为盲人工智能提供声音数据。

此前已有多项关于利用声音的游戏人工智能的研究。Gaina 和 Stephenson [5] 扩展了通用视频游戏人工智能框架，使其支持声音，并训练了一个仅凭声音玩游戏的人工智能。Hegde 等人 [25] 扩展了 VizDoom框架，为人工智能提供游戏内声音，并在多个难度逐渐增加的场景中对其进行训练，以测试其声音感知能力。这些研究结果表明，通过声音学习玩游戏具有潜力。在我们的项目中，DareFightingICE 中的声音将提供给参赛的盲人工智能，以便训练它们仅凭声音玩游戏。

为了便于人工智能获取声音数据，我们扩展了 FightingICE [6] 中的原有接口，新增了方法 *getAudioData*()，用于在游戏的每一帧提供声音数据。每一帧采样的声音数据时长为 16.67 毫秒；请注意，DareFightingICE 与 FightingICE 一样，每秒帧数为 60。在 DareFightingICE 中，立体声以三维格式提供，有助于有效呈现游戏环境。因此，为了让人工智能能够感知声音，并借此定位游戏内对象和游戏内事件，我们提供了双声道格式的声音。

如图2所示，我们不仅在每一帧提供原始数据，还提供两种数据变换（图3）：快速傅里叶变换（FFT）和梅尔频谱图（梅尔频谱）。

- 原始数据：原始音频数据是一个向量 $\mathbf{s} \in \mathbf{R_n}, \mathbf{si} \in [-\mathbf{1}, \mathbf{1}]$ 其中，n 为归一化音频样本的数量。更高的采样频率能够提供更丰富的观测细节。在我们的工作中，采样频率为 48000Hz，从而实现高质量的声音和快速的声音渲染。此外，声音数据的采样时长为 16.67 ms，因此，我们的原始音频数据在左、右两个声道中各包含 800 个样本。需要注意的是，在我们的实现中，我们进行了填充，使每个声道的原始数据均表示为一个长度为 1024 的数组，以提高后续数据变换的计算效率。
- 快速傅里叶变换：我们使用快速傅里叶变换进行第一种变换，将原始数据转换到频域。快速傅里叶变换的输出数据包含两部分：实部和虚部。每部分都是一个长度为1024的数组。
- 梅尔频谱：这类变换已在语音处理领域取得成功 [26]。因此，我们使用短时傅里叶变换（STFT）将原始音频样本转换为频域谱图。短时傅里叶变换是对加窗信号进行的一系列傅里叶变换，其中窗口按给定的帧移移动。随后，我们将梅尔标度应用于这些频率。除窗口大小外，我们采用[26], 中的所有参数，帧移大小为10毫秒，梅尔频率分量为80个。由于一帧的时长短于25毫秒，我们选择1024作为窗口大小。梅尔频谱的输出是一个形状为(80, 3)的向量，其中80为梅尔频带的数量，3为梅尔频谱的数量。

图2展示了提供给人工智能的声音数据结构及各类数据的维度。各维度中的数字 "2" 表示两个声道。我们对左、右声道的音频均进行处理，并将变换后的数据存入两个数组，每个数组存储一次变换的输出。图3展示了使用我们的参数得到的某个动作的音频波形及其变换结果示例。

人工智能使用 Java 或 Python 编写，并且必须实现图 4 所示的 AIInterface 接口。新方法$getAudioData()$在游戏的每一帧提供声音数据。调用此方法时，会向人工智能提供一个 $AudioData$ 对象作为实参。 $AudioData$ 提供对上述声音数据（原始数据、快速傅里叶变换和梅尔频谱）相关信息的访问。

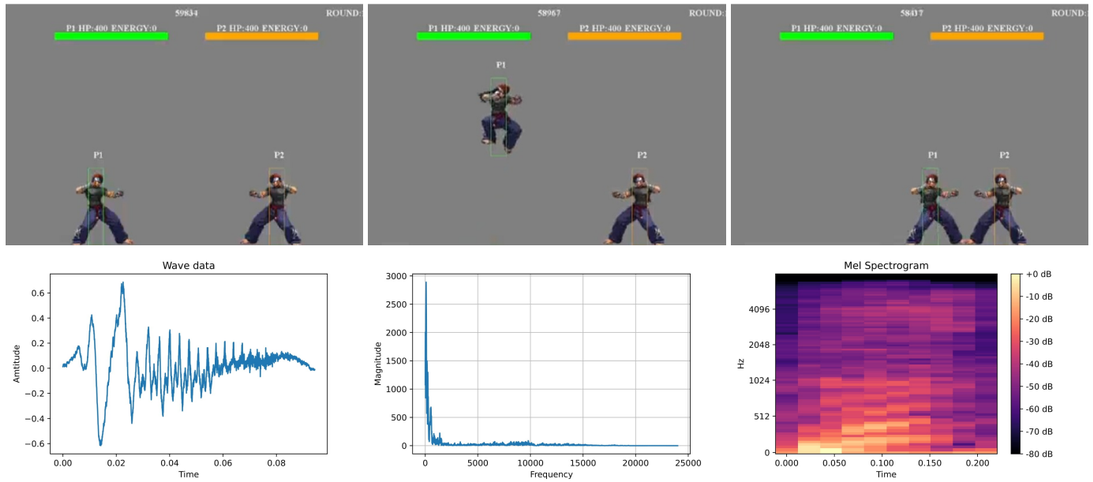


Fig. 3. JUMP action in DareFightingICE. The three in the top show the action sequence of ZEN (the official character in the 2022 competition), the others below show raw sound data, Fourier Transform and Mel-Spectrogram, respectively.

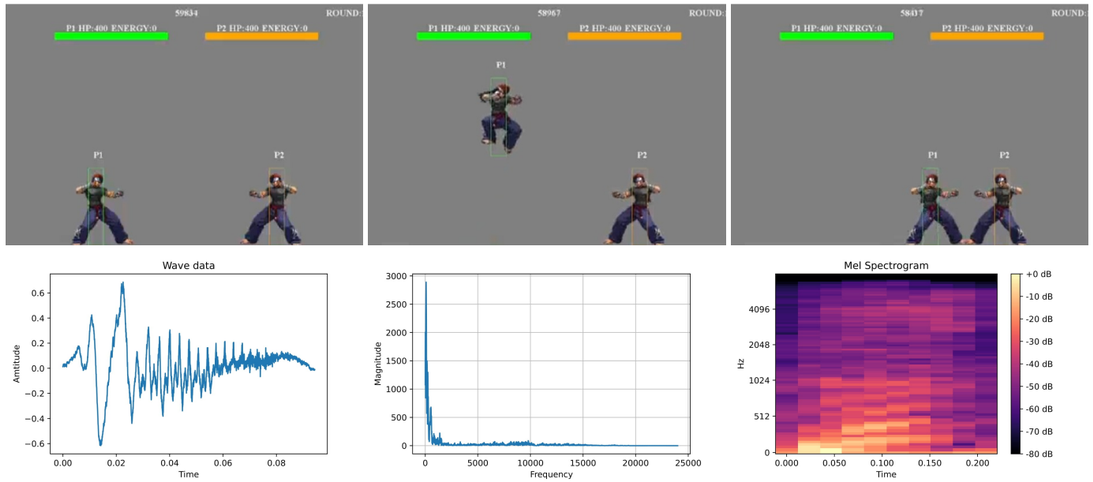


图3. DareFightingICE中的跳跃动作 。上方三幅图展示了ZEN（2022年竞赛的官方角色）的动作序列，其余下方各图分别展示原始声音数据、傅里叶变换和梅尔频谱图。

Our sample blind AI, which uses the interface and sound data provided by DareFightingICE, learns to play the game from sound only. Many Reinforcement Learning techniques have been used to train AIs such as Deep Reinforcement Learning in the work of Mnih et al. [27] and Proximal Policy Optimization [28] used in the AI of Rongqin Liang et al. [29]. Therefore, in our work, we use Reinforcement Learning to train our sample AI. The higher the quality sound design is, the better it represents the game environment, and, therefore, the AI can be expected to better learn the observation provided by the sound data. Thereby, the efficiency of learning progress can be used to evaluate the quality of each submitted sound design. An auxiliary competition paper for CoG 2022 on this AI is under preparation.

```java
package aiinterface;
import struct.*;
/**
 * The interface that defines the methods to
     implement in AI.
 */
public interface AIInterface {
   /**
    * Gets the audio information in each frame.
    * @param ad
    *        the audio information.
    *
    */
   default void getAudioData(AudioData ad){};
   // Other methods which are from the original
       FightingICE platform...
   }
```

Fig. 4. The AIInterface interface

IV. DareFightingICE Competiton

DareFightingICE Competition is a fighting game competition for promoting the sound design of fighting games, targeting VI Players. The competition has two tracks. The first track is a sound-design competition, and the second track is an AI competition. Participants are invited to submit a sound design for VI Players and/or an AI capable of operating with only sound input information.

To achieve synergy between the two tracks, they are uniquely connected with each other. Figure 5 illustrates this point. Namely, the wining sound design of Sound Design Track will be used as the default sound design for AI Track next year. Conversely, the winner AI of AI Track will be used in Sound Design Track instead of the sample blind AI, described in IV-A and in the following. This process will continue as long as the competition is running.

A. Sound Design Track

In this track, participants are asked to make a sound design for DareFightingICE targeting VI Players. They are provided with a fully functional sample sound design and all the sound effects used in the sample, the default sound design of DareFightingICE. They will be allowed to edit a part of the source code in the game as well as add their own sound and any new source code files. This gives them the full ability to make their own sound design.

The participants are allowed to create new sound effects on their own or use already existing sound effects that have the common creative license 0[1]. Apart from creating their own sound effects, participants are allowed to use any techniques to generate sound effects or music for their sound design,

[1]https://creativecommons.org/share-your-work/public-domain/cc0/

我们的示例盲人 AI 使用 DareFightingICE 提供的接口和声音数据，仅通过声音学习玩游戏。许多强化学习技术已被用于训练人工智能，例如 Mnih 等人的研究中使用的深度强化学习 [27] ，以及 Rongqin Liang 等人的人工智能 [29]中使用的近端策略优化 [28] 。因此，在本研究中，我们使用强化学习来训练示例人工智能。声音设计的质量越高，就越能准确地表现游戏环境，因此可以预期，人工智能也能更好地学习声音数据所提供的观测信息。由此，可以利用学习进展的效率来评估各项提交的声音设计的质量。一篇面向 CoG 2022、介绍该人工智能的竞赛配套论文正在撰写中。

```java
package aiinterface;import struct.*;/**定义人
工智能中需要实现的方法的接口。 */public interface
AIInterface {/**获取每一帧的音频信息。 *@param
ad音频信息。 **/defaultvoid
getAudioData(AudioData ad){};// 来自原始
FightingICE平台的其他方法……}
```

图4. AIInterface 接口

IV. DareFightingICE竞赛

DareFightingICE竞赛是一项面向视障玩家的格斗游戏竞赛，旨在推动格斗游戏的声音设计。该竞赛设有两个赛道。第一个赛道是声音设计竞赛，第二个赛道是人工智能竞赛。参赛者可提交面向视障玩家的声音设计和/或能够仅依靠声音输入信息运行的人工智能。

为实现两个赛道之间的协同效应，二者以独特的方式相互关联。图5展示了这一点。具体而言，声音设计赛道的获胜声音设计将作为下一年人工智能赛道的默认声音设计。反过来，人工智能赛道的获胜人工智能将用于声音设计赛道，取代第IV-A节及下文所述的示例盲人 AI。只要竞赛持续举办，这一过程就会延续下去。

A. 声音设计赛道

在此赛道中，参赛者需要为DareFightingICE制作面向视障玩家的声音设计。主办方将提供一套功能完整的示例声音设计，以及该示例中使用的所有音效，即DareFightingICE的默认声音设计。参赛者可以编辑游戏中的部分源代码，也可以添加自己的声音和任何新的源代码文件。这使他们能够完全自主地制作声音设计。

参赛者可以自行创作新音效，也可以使用采用知识共享许可协议的现有音效 0[1]。除自行创作音效外，参赛者还可以使用任何技术，为其声音设计生成音效或音乐，

[1]https://creativecommons.org/share-your-work/public-domain/cc0/

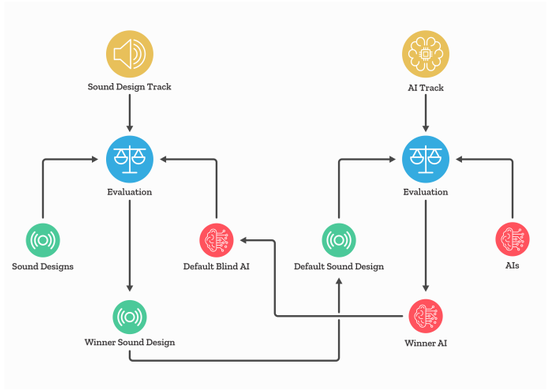


Fig. 5. Competition Work Flow.

including Procedural Content Generation (PCG) [30]. PCG has been used in video games to generate different content including sound effects and music [31]. Giving participants the freedom to use techniques like PCG is our way of making sure they have the resources to create a unique sound design.

Evaluation of sound designs will be done by players with vision while wearing a blindfold. Previous research has shown that when VI Players are not available for evaluation or testing, players with vision with blindfolds can represent them [32] [33] [34]. It was also observed that while the blindfolded players can represent VI Players, they cannot play the game as well as VI Players while being blindfolded. This is because players with vision are not used to playing with no vision [34] [35]. It is for this reason that the blindfolded players will play against a relatively weak AI in the evaluation process, but in future we plan to also add VI judges in the competition.

The total number of players testing a sound design is set to be a least three. If the total number of sound designs is more than five, the sound designs will go through a pre-screening process, due to the time constraint in the competition schedule, and the top five will be selected. They will then go through the screening process. If the number is five or lower, the pre-screening process will be skipped.

Pre-screening:

- Two most capable test players (players with vision) will play against each other for one round using the default sound design. The replay of the same round will be played using all the submitted sound designs – all videos will be of the same match but different sound designs. Then a sound aesthetic survey [36] of those replay videos will be conducted targeting general respondents. The result of this survey will determine the top five sound designs.

Screening:

- The blindfolded players will play against the aforementioned weak AI, a sample Monte-Carlo tree search (MCTS) AI of FightingICE [22]). Note that this weak MCTS AI will play in a non-blind fashion, having access to all the game data provided by the game.
- Each player will play against the AI for three games[2] for each sound design, and the score will be calculated by the HP difference between the player and the AI, in relative to the HP difference when playing without being blindfolded.
- After the play, they (the players) will be asked to complete the same sound aesthetic survey mentioned above. The results of this survey will also be counted in the final score of a sound design of interest.
- In addition, our deep reinforcement learning blind AI will be newly trained with each sound design and then play the game (30 games or 90 rounds per sound design) against the same weak MCTS AI.
- The sample blind AI's learning curve, win-lose ratio, and HP difference will also be used in the total score. In the end, the sound design with the highest overall score will win this track.

B. AI Track

In this track, participants will be asked to make an AI that plays DareFightingICE using only in-game sound data. Participants are allowed to submit their AIs written in Java or Python. The source code of our weak MTCS AI and sample blind DL AI will be provided.

Following the style of the Fighting Game AI Competition run at CIG/CoG during 2014-2021, there are two leagues in this track:

- The Standard League considers the winner of a round as the one with the HP above zero at the time its opponent's HP has reached zero. Both AIs will be given an initial HP of 400. The league is conducted in a round-robin fashion with two games for any pair of entry AIs switching P1 and P2, the players who start from the left side and the right side, respectively. The AI with the highest number of winning rounds becomes the league winner; if necessary, the remaining HPs are used for breaking ties. In this league, our weak sample MTCS AI, playing in the non-blind mode, and our sample blind DL AI, playing in the blind mode, will also participate as guests.
- In the Speedrunning League, the league winner is the AI with the shortest average time to beat both of our aforementioned sample AIs. For each entry AI, five games are conducted with the entry AI being P1 and a sample AI being P2, and another set of five games with the entry AI being P2 and the sample AI being P1. If a sample AI of interest cannot be beaten in 60 seconds, the beating time of its opponent entry AI will be penalized to 70 seconds.

[2]There are three rounds per game, with an initial health point (HP) of 400 and the maximum round time of 60 seconds. Henceforth, unless stated otherwise, this setting is applied to all of the games in this work.

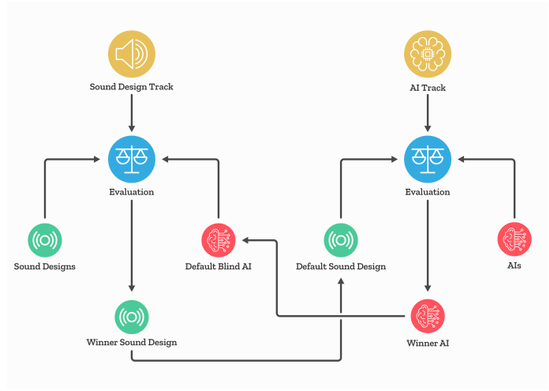


图5. 竞赛流程。

包括程序化内容生成（PCG） [30]。程序化内容生成已被用于电子游戏中，以生成包括音效和音乐在内的各种内容 [31]。我们允许参赛者自由使用程序化内容生成等技术，旨在确保他们拥有打造独特声音设计所需的资源。

声音设计将由佩戴眼罩的健视玩家进行评估。先前研究表明，当没有视障玩家可参与评估或测试时，佩戴眼罩的健视玩家可以代替他们 [32][33] [34]。研究还发现，尽管蒙眼玩家可以代替视障玩家，但他们在蒙眼状态下的游戏表现不及视障玩家。这是因为健视玩家不习惯在没有视觉信息的情况下玩游戏 [34][35]。因此，在评估过程中，蒙眼玩家将与相对较弱的人工智能对战，但我们计划未来也邀请视障评委参与竞赛。

每个声音设计至少由三名玩家进行测试。如果声音设计的总数超过五个，由于竞赛日程的时间限制，将先进行初筛，选出前五名，再进入筛选环节。如果总数不超过五个，则跳过初筛环节。

初筛：

- 两名实力最强的测试玩家（健视玩家）将使用默认声音设计进行一回合对战。随后，将使用所有提交的声音设计播放该回合的回放——所有视频均为同一场对战，仅声音设计不同。接着，将面向普通受访者，针对这些回放视频开展声音美学调查 [36] 。该调查的结果将决定排名前五的声音设计。

筛选：

- 蒙眼玩家将与上述弱人工智能对战，该人工智能是 FightingICE 的一个蒙特卡洛树搜索（MCTS）人工智能示例 [22]）。请注意，这个较弱的蒙特卡洛树搜索人工智能将以非蒙眼方式进行游戏，可以访问游戏提供的所有游戏数据。
- 针对每种声音设计，每位玩家将与人工智能对战三局[2]，得分将根据玩家与人工智能之间的生命值差相对于未佩戴眼罩时生命值差的变化来计算。
- 对战结束后，玩家将被要求完成上述同一份声音美学调查。该调查的结果也将计入相应声音设计的最终得分。
- 此外，我们的深度强化学习盲人 AI 将针对每种声音设计重新训练，然后与同一个较弱的蒙特卡洛树搜索人工智能进行对战（每种声音设计进行 30 局，即 90 回合）。
- 示例盲人 AI 的学习曲线、胜负比和生命值差也将计入总分。最终，总分最高的声音设计将赢得本赛道的冠军。

B. 人工智能赛道

在本赛道中，参赛者需要开发一个仅使用游戏内声音数据来游玩 DareFightingICE 的人工智能。参赛者可以提交用 Java 或 Python 编写的人工智能。我们将提供较弱的 MTCS人工智能和示例盲模式深度学习人工智能的源代码。

沿用2014—2021年期间在 CIG/CoG 举办的格斗游戏人工智能竞赛的形式，本赛道设有两个联赛：

- 标准联赛将一回合中对手生命值降至零时自身生命值仍高于零的一方判为获胜者。双方人工智能的初始生命值均为400。该联赛采用循环赛制，每对参赛人工智能进行两场比赛，并交换 P1 和 P2 的位置，二者分别为从左侧和右侧开始比赛的玩家。获胜回合数最多的人工智能将成为联赛冠军；必要时，将根据剩余生命值打破平局。在本联赛中，我们较弱的示例 MTCS人工智能将以非盲模式运行，示例盲模式深度学习人工智能将以盲模式运行，二者也将作为嘉宾参赛。
- 在速通联赛中，击败上述两个示例人工智能所用时间的平均值最短的人工智能将成为联赛冠军。对于每个参赛人工智能，将进行五场由参赛人工智能担任 P1、示例人工智能担任 P2 的比赛，再进行五场由参赛人工智能担任 P2、示例人工智能担任 P1 的比赛。如果参赛人工智能未能在 60 秒内击败某个示例人工智能，则其击败该示例人工智能的用时将被罚记为 70 秒。

[2]每场比赛共三回合，初始生命值（HP）为400，每回合最大时限为60秒。下文中，除非另有说明，本文所有比赛均采用此设置。

C. Environments

In DareFightingICE Competition, a single environment will be used to run both tracks. More specifically, several computers will be used that have the same specification, i.e., CPU: Intel(R) Xeon(R) W-2135 CPU@ 3.70GHz 3.70 GHz, RAM: 16 GB, GPU: NVIDIA Quadro P1000 4GB VRAM, and OS: Windows 10. These computers will be deployed to evaluate sound designs submitted in Sound Design Track and to assess AIs submitted in AI Track.

V. AI Selection For DareFightingICE

This section describes AI vs. AI and Human vs. AI experiments. In the AI vs. AI experiment, three MCTS AIs with different parameter settings are used to find a suitable opponent AI for not only as an opponent AI fighting against the blindfolded players in Sound Design Track but also as one of the guest AIs in AI Track. These three MCTS AI named MctsAI165, MctsAI115, and MctsAI65 are the same as a sample MCTS AI[3] from FightingICE, but with the MCTS execution time [37] set to 16.5 ms (the original setting), 11.5 ms, and 6.5 ms, respectively. The execution time sets the time budget of the MCTS algorithm in use, and hence the strength of MCTS will theoretically drop when the execution time decreases.

A. AI vs. AI

This experiment evaluates performance among the three MCTS AIs. Each MCTS AI played 50 games against each of the other two AIs, switching sides between P1 and P2 every 25 games. The ratio of the number of winning rounds over 300 rounds is then used to evaluate the performance of each MCTS AI. The winning ratios of MctsAI165, MctsAI115, and MctsAI65 are 0.96, 0.49, and 0.05, respectively.

B. Human vs. AI

The experiment here examines human players' performance against the three MCTS AIs in normal play (without being blindfolded) and against the selected MCTS AI in blindfolded play. Three human players participated in this experiment and they played against the three MCTS AIs in different orders, according to the Latin square order. The human players always played as P1 three games against a given opponent AI.

We define δ (1) to measure the normalized performance of a player of interest against an opponent AI, where d is given in (2), and due to the initial setting of HP, min and max are -400 and 400, respectively. A value near 0.5 of δ indicates that both player and AI have a similar fighting performance. Finally, we define "Performance Retention Rate (PRR)" in (3) to show how much the players' performance is retained in blindfold play.

$$\delta = \frac{d - min}{max - min} \tag{1}$$

$$d = \mathrm{HP}_{\mathrm{Player}} - \mathrm{HP}_{\mathrm{AI}} \tag{2}$$

[3]http://www.ice.ci.ritsumei.ac.jp/ ftgaic/Downloadfiles/MctsAi.zip

TABLE II
Normalized performance of each player in normal play against different AIs, where x and y in x (y) show the average and standard deviation, respectively

Player \ AI	MctsAI165	MctsAI115	MctsAI65
Player 1	0.22 (0.11)	0.34 (0.11)	0.43 (0.15)
Player 2	0.23 (0.08)	0.36 (0.11)	0.59 (0.08)
Player 3	0.44 (0.12)	0.46 (0.16)	0.51 (0.21)
Average	0.30	0.39	0.51

TABLE III
Normalized performance and PRR of each player in blindfold play against the selected MctsAI65, where x and y in x (y) show the average and standard deviation, respectively

Player	δ	PRR
Player 1	0.26 (0.09)	0.59
Player 2	0.46 (0.05)	0.78
Player 3	0.38 (0.12)	0.74
Average	0.36	0.71

$$\mathrm{PRR} = \frac{\mathrm{average}(\delta_{\mathrm{blind}})}{\mathrm{average}(\delta_{\mathrm{normal}})} \tag{3}$$

Table II shows that the human players have a higher performance when the execution time of MCTS AI decreases in normal play. On average, they have close fights with MctsAI65, as shown by 0.51 on the bottom-right cell. Since fighting against a too strong opponent AI such as the other two AIs would make the resulting PRR unreliable, MctsAI65 is selected as the aforementioned weak MCTS AI.

Table III shows that on average PRR against MctsAI65 is 0.71. This indicates that the players have a mild performance drop and can keep some of their playing skills with the default sound design[4]. However, a better sound design should lead to a higher value of PRR. As a result, PRR is used as one of the metrics to evaluate sound designs in Sound Design Track.

VI. Conclusions and Future Work

In this paper, we introduced our new DareFightingICE Competition which has two tracks: Sound Design Track and AI Track. We also outlined the format and the motivation behind this competition. The competition is set to run at CoG 2022. The platform for the competition is DarefightingICE – an enhanced version of FightingICE – with an enhanced sound design and a new AI interface that can give sound data to AIs.

For evaluation of submissions to this competition, we conducted an experiment to find the best suited opponent AI that our blindfolded human players will play against in Sound Design Track. The experiment was divided into two parts: AI vs AI and AI vs Human. Different variations of an MCTS AI were used, and it was found that MctsAI65 was best suited for the evaluation process of this track. Playing against MctsAI65 in blindfold play led to an average PRR of 0.71, and we

[4]Video showing a game in blind play: https://tinyurl.com/cog2022blind

C. 环境

DareFightingICECompetition 的两个赛道将使用统一的运行环境。具体而言，将使用多台配置相同的计算机，即中央处理器：英特尔至强 W-2135，主频 3.70GHz 3.70 GHz；随机存取存储器：16 GB；图形处理器：NVIDIA Quadro P1000，配备 4GB 显存；操作系统：Windows 10。这些计算机将用于评估声音设计赛道提交的声音设计，以及人工智能赛道提交的人工智能。

V. DareFightingICE 的人工智能选择

本节介绍人工智能对战人工智能和人类对战人工智能的实验。在人工智能对战人工智能实验中，我们使用三种参数设置不同的蒙特卡洛树搜索人工智能，以寻找合适的对手人工智能，既用于在声音设计赛道中与蒙眼玩家对战，也作为人工智能赛道中的嘉宾人工智能之一。这三种蒙特卡洛树搜索人工智能分别名为 MctsAI165、MctsAI115 和 MctsAI65，与 FightingICE 中的示例 MCTSAI[3] 相同，但其蒙特卡洛树搜索执行时间 [37] 分别设为 16.5 毫秒（原始设置）、11.5 毫秒和 6.5 毫秒。执行时间规定了所用蒙特卡洛树搜索算法的时间预算，因此，理论上，蒙特卡洛树搜索的实力会随着执行时间的缩短而下降。

A. 人工智能对战人工智能

本实验评估三种蒙特卡洛树搜索人工智能的性能。每种蒙特卡洛树搜索人工智能分别与另外两种人工智能各进行 50 场游戏，每 25 场在 P1 和 P2 之间交换一次位置。随后，以 300 个回合中的获胜回合数占比来评估每种蒙特卡洛树搜索人工智能的性能。MctsAI165、MctsAI115 和 MctsAI65 的胜率分别为 0.96、0.49 和 0.05。

B. 人类对战人工智能

本实验考察了人类玩家在正常游玩（不戴眼罩）时对战三种蒙特卡洛树搜索人工智能的表现，以及在蒙眼游戏时对战选定的蒙特卡洛树搜索人工智能的表现。三名人类玩家参与了本实验，并按照拉丁方顺序，以不同次序对战这三种蒙特卡洛树搜索人工智能。人类玩家始终作为 P1，与每个给定的对手人工智能进行三场游戏。

我们在式（1）中定义了 δ，用于衡量所考察玩家对战某个对手人工智能时的归一化表现，其中 d 由式（2）给出；根据生命值的初始设置，min 和 max 分别为 -400 和 400。δ 的值接近 0.5，表明玩家与人工智能的格斗表现相近。最后，我们在式（3）中定义了“表现保留率（PRR）”，用于表示玩家在蒙眼游玩时保留了多少表现水平。

$$\delta = \frac{d - min}{max - min} \tag{1}$$

$$d = \mathrm{HP}_{\mathrm{Player}} - \mathrm{HP}_{\mathrm{AI}} \tag{2}$$

[3]http://www.ice.ci.ritsumei.ac.jp/ ftgaic/Downloadfiles/MctsAi.zip

表 II 各玩家在正常游玩时对战不同人工智能的归一化表现，其中 X(Y) 中的 X 和 Y 分别表示平均值和标准差

玩家 \ AI	MctsAI165	MctsAI115	MctsAI65
玩家 1	0.22 (0.11)	0.34 (0.11)	0.43 (0.15)
玩家 2	0.23 (0.08)	0.36 (0.11)	0.59 (0.08)
玩家 3	0.44 (0.12)	0.46 (0.16)	0.51 (0.21)
平均值	0.30	0.39	0.51

表 III 各玩家在蒙眼游玩中对抗所选 MctsAI65 时的归一化表现和 PRR，其中 X(Y) 中的 X 和 Y 分别表示平均值和标准差

玩家	δ	PRR
玩家 1	0.26 (0.09)	0.59
玩家 2	0.46 (0.05)	0.78
玩家 3	0.38 (0.12)	0.74
平均值	0.36	0.71

$$\mathrm{PRR} = \frac{\mathrm{average}(\delta_{\mathrm{blind}})}{\mathrm{average}(\delta_{\mathrm{normal}})} \tag{3}$$

表 II 显示，在正常游玩中，随着蒙特卡洛树搜索人工智能的执行时间缩短，人类玩家的表现有所提升。从平均值来看，他们与 MctsAI65 势均力敌，右下角单元格中的 0.51 就体现了这一点。由于与过于强大的对手人工智能（例如另外两个人工智能）对战会使得到的 PRR 不可靠，因此选择 MctsAI65 作为前述较弱的蒙特卡洛树搜索人工智能。

表III显示，对阵MctsAI65时，PRR的平均值为 0.71。这表明，使用默认声音设计时，玩家的表现略有下降，但仍能保留部分游戏技能[4]。然而，更好的声音设计应能带来更高的PRR值。因此，PRR被用作声音设计赛道中评估声音设计的指标之一。

VI. 结论与未来工作

本文介绍了我们全新的DareFightingICE竞赛，该竞赛设有两个赛道：声音设计赛道和人工智能赛道。我们还概述了竞赛的形式及其背后的动机。该竞赛定于CoG 2022期间举行。竞赛平台为DarefightingICE，它是FightingICE的增强版，具有增强的声音设计和可向人工智能提供声音数据的新人工智能接口。

为了评估本次竞赛的参赛作品，我们开展了一项实验，以找出最适合在声音设计赛道中与蒙眼人类玩家对战的对手人工智能。实验分为两部分：人工智能对战人工智能和人工智能对战人类。实验使用了蒙特卡洛树搜索人工智能的不同变体，结果发现 MctsAI65 最适合用于该赛道的评估。在蒙眼游玩中与 MctsAI65 对战时，PRR 的平均值为 0.71，而我们

[4]展示盲玩对局的视频：https://tinyurl.com/cog2022blind

perceived that a better sound design should at least have a PRR higher than 0.71.

In the future, we plan to strengthen our sample blind DL AI by taking a new or different approach, such as multimodal transfer reinforcement learning [38]. We also plan to create a sound design using PCG for DareFightingICE and to continue running DareFightingICE Competition at subsequent CoG for a number of years to ensure that our participants get enough time and resources to come up with better and better solutions. Finally, findings from this competition will be regularly summarized and reported by us at subsequent CoG or journals, such as IEEE Transactions on Games.

REFERENCES

[1] J. Clement, "Video game industry - Statistics & Facts," 2021. [Online]. Available: https://www.statista.com/topics/868/video-games/#dossier-chapter1. [Accessed: 28 Feb., 2022].

[2] G. N. Yannakakis, N. Georgios and J. Togelius, *Artificial intelligence and games*, New York: Springer, Vol. 2, 2018.

[3] Q. Yin, J. Yang, W. Ni, B. Liang, and K. Huang, "AI in Games: Techniques, Challenges and Opportunities," arXiv preprint arXiv:2111.07631, 2021.

[4] I. Khaliq, and I. D. Torre, "A study on accessibility in games for the visually impaired," in *Proceedings of the 5th EAI International Conference on Smart Objects and Technologies for Social Good*, pp. 142-148, Sep. 2019.

[5] R. D. Gaina and M. Stephenson, ""Did You Hear That?" Learning to Play Video Games from Audio Cues," in *Proceedings of the 2019 IEEE Conference on Games (CoG)*, pp. 1-4, 2019.

[6] Feiyu Lu, Kaito Yamamoto, Luis H. Nomura, Syunsuke Mizuno, Young-Min Lee, and Ruck Thawonmas, "Fighting Game Artificial Intelligence Competition Platform," Proc. of the 2nd IEEE Global Conference on Consumer Electronics (GCCE 2013), Tokyo, Japan, pp. 320-323, Oct. 1-4, 2013. (https://www.ice.ci.ritsumei.ac.jp/-ftgaic/index.htm)

[7] K. Tuuri, O. Koskela, J. Vahlo, and H. Tissari, "Identifying the Impact of Game Music both Within and Beyond Gameplay," in *International Conference on Entertainment Computing, Springer, Cham*, pp. 411-418, Nov. 2021.

[8] J. Byun, and C. S. Loh, "Audial engagement: Effects of game sound on learner engagement in digital game-based learning environments," in *Computers in Human Behavior*, vol.46, pp.129-138, 2015.

[9] J. Zhang, and X. Fu, "The influence of background music of video games on immersion," in *Journal of Psychology and Psychotherapy*, vol. 5, no. 4, p.1, 2015.

[10] F. Andersen, C. L. King, and A. A. Gunawan, "Audio Influence on Game Atmosphere during Various Game Events," in *Procedia Computer Science*, vol.179, pp.222-231, 2021.

[11] C. Johanson, and R. L. Mandryk, "Scaffolding player location awareness through audio cues in first-person shooters," in *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 3450-3461, May 2016.

[12] P. Ng, and K. Nesbitt, "Informative sound design in video games," in *Proceedings of the 9th Australasian conference on interactive entertainment: matters of life and death*, pp. 1-9, Sep. 2013.

[13] R. Andrade, M. J. Rogerson, J. Waycott, S. Baker, and F. Vetere, "Playing blind: Revealing the world of gamers with visual impairment," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, pp. 1-14, May 2019.

[14] C. Chai, B. T. Lau, and Z. Pan, "Hungry Cat—a serious game for conveying spatial information to the visually impaired," in *Multimodal Technologies and Interaction*, vol. 3, no. 1, p.12, 2019.

[15] J. Walton-Rivers, P. R. Williams and R. Bartle, "The 2018 Hanabi competition," in *Proceedings of the 2019 IEEE Conference on Games (CoG)*, pp. 1-8, 2019.

[16] N. Justesen, L. M. Uth, C. Jakobsen, P. D. Moore, J. Togelius and S. Risi, "Blood Bowl: A New Board Game Challenge and Competition for AI," in *Proceedings of the 2019 IEEE Conference on Games (CoG)*, pp. 1-8, 2019.

[17] S. Reis, L. P. Reis and N. Lau, "VGC AI Competition - A New Model of Meta-Game Balance AI Competition," in *Proceedings of the 2021 IEEE Conference on Games (CoG)*, pp. 01-08, 2021.

[18] Q. Tyrell Davis, "Carle's Game: An Open-Ended Challenge in Exploratory Machine Creativity," in *Proceedings of the 2021 IEEE Conference on Games (CoG)*, pp. 01-08, 2021.

[19] Game Music and Sound Design Conference, 2018. [online] Available at: https://www.gamesoundcon.com/ [Accessed: 28 Feb., 2022].

[20] The Berlin International Sound Design Competition, 2018. [online] Available at: https://www.bifsc.org/sound-design-competition/ [Accessed: 28 Feb., 2022].

[21] Film Music Contest's Music for video game, 2021. [online] Available at: https://www.fmcontest.com/music-for-video-game/ [Accessed: 28 Feb., 2022].

[22] R. Ishii, K. Fujimaki and R. Thawonmas, "Fighting-Game Gameplay Generation Using Highlight Cues," in *IEEE Transactions on Games*, doi: 10.1109/TG.2021.3097071, 2021.

[23] T. Chen, F. Richoux, J. M. Torres and K. Inoue, "Interpretable Utility-based Models Applied to the FightingICE Platform," in *Proceedings of the 2021 IEEE Conference on Games (CoG)*, pp. 1-8, 2021.

[24] Z. Tang, Y. Zhu, D. Zhao and S. M. Lucas, "Enhanced Rolling Horizon Evolution Algorithm with Opponent Model Learning," in *IEEE Transactions on Games*, doi: 10.1109/TG.2020.3022698, 2020.

[25] S. Hegde, A. Kanervisto and A. Petrenko, "Agents that Listen: High-Throughput Reinforcement Learning with Multiple Sensory Systems," in *Proceedings of the 2021 IEEE Conference on Games (CoG)*, pp. 1-5, 2021.

[26] D. Garcia-Romero, G. Sell, and A. Mccree, "Magneto: X-vector magnitude estimation network plus offset for improved speaker recognition," in *Proceedings of the Odyssey 2020 the speaker and language recognition workshop*, pp. 1-8, May 2020.

[27] V. Mnih, K. Kavukcuoglu, D. Silver, A. Graves, I. Antonoglou, D. Wierstra, and M. Riedmiller, "Playing atari with deep reinforcement learning," in *NIPS Deep Learning Workshop*, 9 pages, 2013.

[28] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," arXiv preprint arXiv:1707.06347, 2017.

[29] R. Liang, Y. Zhu, Z. Tang, M. Yang and X. Zhu, "Proximal Policy Optimization with Elo-based Opponent Selection and Combination with Enhanced Rolling Horizon Evolution Algorithm," in *Proceedings of the 2021 IEEE Conference on Games (CoG)*, pp. 1-4, 2021.

[30] N. Shaker, J. Togelius, and M. J. Nelson, *Procedural content generation in games*, Switzerland: Springer International Publishing, 2016.

[31] J. Liu, S. Snodgrass, A. Khalifa, S. Risi, G. N. Yannakakis, and J. Togelius, "Deep learning for procedural content generation. Neural Computing and Applications," in *Proceedings of the Neural Comput and Applications*, vol. 33, no, 1, pp. 19–37, 2021.

[32] J. F. P. Cheiran, L. Nedel and M. S. Pimenta, "Inclusive Games: A Multimodal Experience for Blind Players," in *Proceedings of the 2011 Brazilian Symposium on Games and Digital Entertainment*, pp. 164-172, 2011.

[33] A. Adkins, K. Kohm, R. Zhang, and N. Gustafson, "Lost in Spaze: An Audio Maze Game for the Visually Impaired," in *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*, pp. 1-6, April 2020.

[34] N. Mohd Norowi, H. Azman, and N. Wahiza Abdul Wahat, "Apple Swipe: A Mobile Game Apps for Visually Impaired Users Using Binaural Sounds," in *Asian CHI Symposium*, pp. 196-201, May 2021.

[35] A. M. Silverman, "The perils of playing blind: Problems with blindness simulation and a better way to teach about blindness," in *Journal of Blindness Innovation and Research*, vol. 5, no. 2, 2015.

[36] M. H. Phan, J. R. Keebler, and B. S. Chaparro, "The development and validation of the game user experience satisfaction scale (GUESS)," Human factors, vol. 58, no. 8, pp. 1217-1247, 2016.

[37] M. Ishihara, T. Miyazaki, C. Y. Chu, T. Harada, and R. Thawonmas, "Applying and improving Monte-Carlo Tree Search in a fighting game AI," in *Proceedings of the 13th international conference on advances in computer entertainment technology*, pp. 1-6, Nov. 2016.

[38] Silva, Rui, Miguel Vasco, Francisco S. Melo, Ana Paiva, and Manuela Veloso."Playing Games in the Dark: An Approach for Cross-Modality Transfer in Reinforcement Learning." *Proceedings of the 19th International Conference on Autonomous Agents and MultiAgent Systems*, pp. 1260-1268. 2020.

认为，更好的声音设计至少应使 PRR 高于 0.71。

未来，我们计划采用新的或不同的方法，例如多模态迁移强化学习 [38]，来增强我们的示例盲模式深度学习人工智能。我们还计划使用程序化内容生成为 DareFightingICE进行声音设计，并在未来数年的CoG上持续举办DareFightingICE竞赛，以确保参赛者有足够的时间和资源提出越来越好的解决方案。最后，我们将定期总结本竞赛的研究成果，并在后续的CoG或《IEEE游戏汇刊》等期刊上进行报告。

参考文献

[1] J. Clement，“电子游戏产业——统计数据与事实”，2021年。[在线]。网址：https://www.statista.com/topics/868/video-games/#dossier- chapter1。 [访问日期：2月28日， 2022]。[2] G. N. Yannakakis、N. Georgios和J. Togelius，人工智能与游戏，纽约：施普林格，第2卷，2018年。[3] Q. Yin、J. Yang、W. Ni、B. Liang和K. Huang，“游戏中的人工智能：技术、挑战与机遇”，arXiv预印本arXiv:2111 07631，2021年。[4] I. Khaliq和I. D. Torre，“面向视障人士的游戏无障碍性研究”，载于第五届*EAI*社会公益智能物体与技术国际会议论文集，第142-148页，2019年9月。[5] R. D. Gaina和M. Stephenson，“‘你听到了吗？’通过音频提示学习玩电子游戏”，载于*2019*年*IEEE*游戏大会（*CoG*）论文集，第1-4页，2019年。[6] Feiyu Lu、Kaito Yamamoto、Luis H. Nomura、Syunsuke Mizuno、Young-Min Lee和Ruck Thawonmas，“格斗游戏人工智能竞赛平台”，第二届IEEE全球消费电子大会（GCCE 2013）论文集，日本东京，第320-323页，2013年10月1日至4日。（https://www.ice.ci.ritsumei.ac.jp/-ftgaic/index.htm）[7] K. Tuuri、O. Koskela、J. Vahlo和H. Tissari，“识别游戏音乐在游戏玩法内外的影响”，载于娱乐计算国际会议，施普林格，查姆，第411-418页，2021年11月。[8] J. Byun和C. S. Loh，“听觉投入：游戏声音对数字游戏化学习环境中学习者投入的影响”，载于人类行为中的计算机，第46卷，第129-138页，2015年。[9] J. Zhang和X. Fu，“电子游戏背景音乐对沉浸感的影响”，载于心理学与心理治疗杂志，第5卷，第4期，第1页，2015年。[10] F. Andersen、C. L. King和A. A. Gunawan，“不同游戏事件中音频对游戏氛围的影响”，载于计算机科学程序，第179卷，第222-231页，2021年。[11] C. Johanson和R. L. Mandryk，“通过音频提示辅助第一人称射击游戏中的玩家位置感知”，载于*2016*年*CHI*计算系统人因会议论文集，第3450-3461页，2016年5月。[12] P. Ng和K. Nesbitt，“电子游戏中的信息传达型声音设计”，载于第九届澳大拉西亚互动娱乐会议论文集：生死攸关，第1-9页，2013年9月。[13] R. Andrade、M. J. Rogerson、J. Waycott、S. Baker和F. Vetere，“盲玩：揭示视觉障碍玩家的世界”，载于*2019*年*CHI*计算系统人因会议论文集，第1-14页，2019年5月。[14] C. Chai、B. T. Lau和Z. Pan，“Hungry Cat——一款向视障人士传达空间信息的严肃游戏”，载于多模态技术与交互，第3卷，第1期，第12页，2019年。[15] J. Walton-Rivers、P. R. Williams和R. Bartle，“2018年花火竞赛”，载于*2019*年*IEEE*游戏大会（*CoG*）论文集，第1-8页，2019年。[16] N. Justesen、L. M. Uth、C. Jakobsen、P. D. Moore、J. Togelius和S. Risi，“血腥橄榄球：面向人工智能的新桌面游戏挑战与竞赛”，载于*2019*年*IEEE*游戏大会（*CoG*）论文集，第1-8页，2019年。

[17] S. Reis、L. P. Reis和N. Lau，“VGC人工智能竞赛——一种元游戏平衡人工智能竞赛新模式”，载于2021年IEEE游戏会议（CoG）论文集，第01-08页，2021年。[18] Q. Tyrell Davis，“Carle的游戏：探索性机器创造力的开放式挑战”，载于2021年IEEE游戏会议（CoG）论文集，第01-08页，2021年。[19] 游戏音乐与声音设计会议，2018年。 [在线]网址：https://www.gamesoundcon.com/ [访问日期：2月28日，2022]。[20] 柏林国际声音设计大赛，2018年。 [在线] 网址：https://www.bifsc.org/sound-design-competition/ [访问日期：2月28日， 2022]。[21] 电影音乐大赛的电子游戏音乐，2021年。 [在线]网址：https://www.fmcontest.com/music-for-video-game/ [访问日期：2月28日，2022]。[22] R. Ishii、K. Fujimaki和R. Thawonmas，“利用精彩片段线索进行格斗游戏玩法生成”，载于IEEE游戏汇刊，doi: 10.1109/TG.2021.3097071，2021年。[23] T. Chen、F. Richoux、J. M. Torres和K. Inoue，“应用于FightingICE平台的可解释效用模型”，载于2021年IEEE游戏会议（CoG）论文集，第1-8页，2021年。[24] Z. Tang、Y. Zhu、D. Zhao和S. M. Lucas，“结合对手模型学习的增强型滚动时域进化算法”，载于IEEE游戏汇刊，doi: 10.1109/TG.2020.3022698，2020年。[25] S. Hegde、A. Kanervisto和A. Petrenko，“会倾听的智能体：利用多感官系统实现高吞吐量强化学习”，载于2021年5，2021年1 IEEE游戏会议（CoG）论文集，第-页。[26] D. Garcia-Romero、G. Sell和A. Mccree，“Magneto：结合X向量幅值估计网络与偏移量以改进说话人识别”，载于Odyssey 2020说话人与语言识别研讨会论文集，第1-8页，2020年5月。[27] V. Mnih、K. Kavukcuoglu、D. Silver、A. Graves、I. Antonoglou、D. Wierstra和M. Riedmiller，“通过深度强化学习玩雅达利游戏”，载于NIPS深度学习研讨会，9页，2013年。[28] J. Schulman、F. Wolski、P. Dhariwal、A. Radford和O. Klimov，“近端策略优化算法”，arXiv预印本arXiv:1707 06347，2017年。[29] R. Liang、Y. Zhu、Z. Tang、M. Yang和X. Zhu，“采用基于Elo等级分的对手选择并结合增强型滚动时域进化算法的近端策略优化”，载于2021年IEEE游戏会议（CoG）论文集，第1-4页，2021年。[30] N. Shaker、J. Togelius和M. J. Nelson，游戏中的程序化内容生成，瑞士：施普林格国际出版公司，2016年。[31] J. Liu、S. Snodgrass、A. Khalifa、S. Risi、G. N. Yannakakis和J. Togelius，“用于程序化内容生成的深度学习。神经计算与应用”，载于神经计算与应用论文集，第33卷，第1期，第19–37页，2021年。[32] J. F. P. Cheiran、L. Nedel和M. S. Pimenta，“包容性游戏：面向盲人玩家的多模态体验”，载于2011年巴西游戏与数字娱乐研讨会论文集，第164 172页，2011年-。[33] A. Adkins、K. Kohm、R. Zhang和N. Gustafson，“Lost in Spaze：一款面向视障人士的音频迷宫游戏”，载于2020年CHI计算系统人因会议扩展摘要集，第1-6页，2020年4月。[34] N. Mohd Norowi、H. Azman和N. Wahiza Abdul Wahat，“Apple Swipe：一款面向视障用户、采用双耳声音的移动游戏应用”，载于亚洲CHI研讨会，第196-201页，2021年5月。[35] A. M. Silverman，“盲玩的风险：失明模拟存在的问题及开展失明教育的更佳方式”，载于失明创新与研究杂志，第5卷，第2期，2015年。[36] M. H. Phan、J. R. Keebler和B. S. Chaparro，“游戏用户体验满意度量表（GUESS）的开发与验证”，人因学，第58卷，第8期，第1217-1247页，2016年。[37] M. Ishihara、T. Miyazaki、C. Y. Chu、T. Harada和R. Thawonmas，“在格斗游戏人工智能中应用并改进蒙特卡洛树搜索”，载于第13届计算机娱乐技术进展国际会议论文集，第1-6页，2016年11月。[38] Silva, Rui、Miguel Vasco、Francisco S. Melo、Ana Paiva和Manuela Veloso。“在黑暗中玩游戏：一种强化学习跨模态迁移方法。”第19届自主智能体与多智能体系统国际会议论文集，第1260-1268页。2020年。